DEGRADATION-AWARE MODEL PREDICTIVE CONTROL FOR A FREIBURG RENEWABLE DISTRICT MICROGRID

with AC Grid Validation, Optional OpenDSS Unbalanced Feeder Checking and Rainflow Battery Aging Analysis

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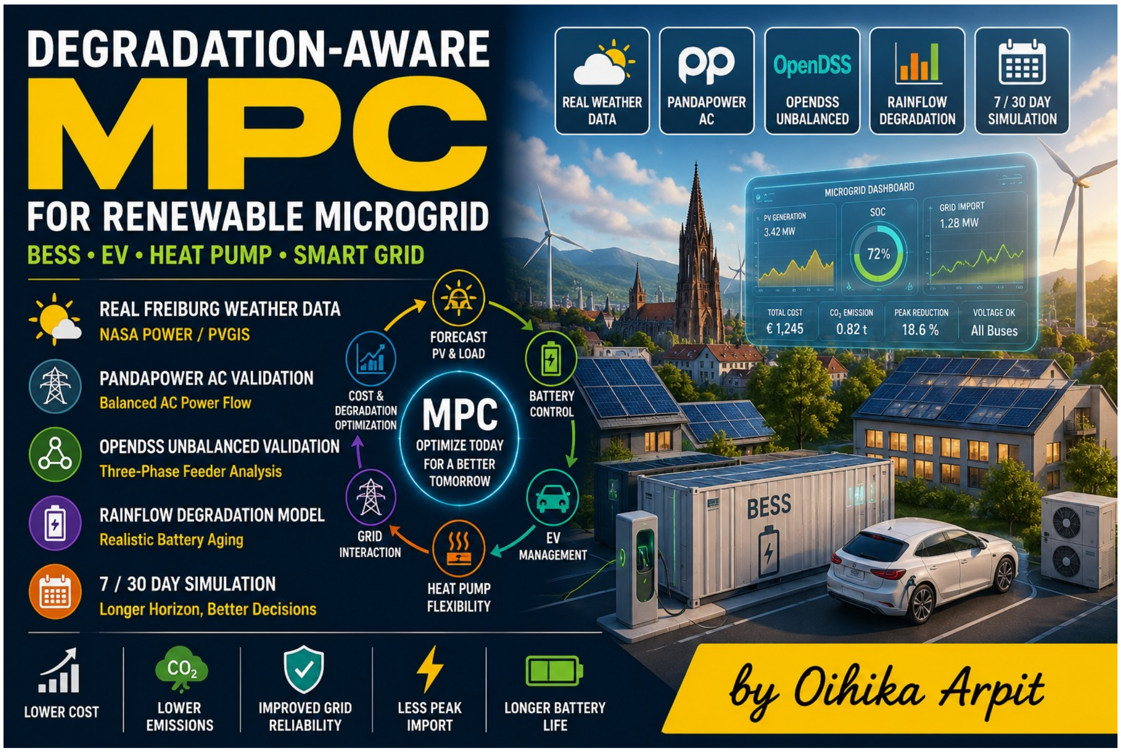
Repository Name: degradation-aware-mpc-freiburg-microgrid



Project thumbnail: Freiburg renewable district microgrid concept with PV, BESS, EV charging and smart-grid control.

IQ 100X Technical Project Report | Renewable Energy Systems | Smart Grid Control | Battery Degradation-Aware Optimization

ABSTRACT

This project presents a degradation-aware Model Predictive Control (MPC) framework for a representative renewable district microgrid in Freiburg, Germany. The system coordinates photovoltaic generation, wind generation, battery energy storage, electric vehicle charging, heat-pump flexibility, dynamic tariff response, carbon-aware dispatch, feeder constraints and grid-supportive battery operation. The project is designed as a research-grade applied simulation rather than a simple dashboard. It includes real-weather-oriented Freiburg renewable input, a 7-day or 30-day simulation option, balanced AC validation using pandapower, optional unbalanced three-phase checking using OpenDSS and rainflow-based battery aging analysis. The IQ 100X design objective is to make the work admission-ready for German MSc programs by combining renewable energy engineering, control theory, grid validation, energy storage aging and professional output visualization.

Keywords - Model Predictive Control, MPC, Microgrid, Freiburg, PV, BESS, EV Charging, Heat Pump, pandapower, OpenDSS, Rainflow Degradation, Smart Grid, Renewable Energy.

I. INTRODUCTION

Renewable-energy-dominated distribution systems require intelligent control methods that can handle variable generation, flexible demand, grid limits and battery degradation. A simple rule-based controller may charge a battery during solar surplus and discharge it during evening peak, but it does not systematically consider future solar forecasts, electricity price, carbon intensity, state of charge, feeder congestion or voltage constraints. This project develops a degradation-aware MPC system for a Freiburg renewable district microgrid to address these limitations.

The central research question is: how can predictive control coordinate PV, wind, BESS, EV charging and heat-pump flexibility to reduce operating cost, emissions, peak import, feeder stress and battery aging while maintaining acceptable distribution-network behavior?

II. IQ 100X CONTRIBUTIONS

- Real-weather-oriented Freiburg renewable input using NASA POWER/PVGIS-style irradiance, temperature and wind signals.
- Extended simulation horizon from a small 48-hour case to 7-day or 30-day comparison.
- Degradation-aware MPC that penalizes battery cycling and manages BESS state of charge.
- pandapower AC validation layer for balanced distribution-network voltage and line-loading checks.
- Optional OpenDSS unbalanced three-phase feeder validation for stronger grid-engineering credibility.
- Rainflow battery aging assessment using SOC cycle depth rather than only total throughput.
- SCADA-style outputs including alarms, KPI dashboards, Excel results and HTML visualizations.

III. SYSTEM ARCHITECTURE

The modeled district contains representative residential and commercial demand, heat-pump load, EV charging flexibility, PV generation, wind generation and a grid interconnection. The MPC controller receives forecasts of demand, renewable generation, price and carbon intensity. It then calculates hourly control actions for battery charge/discharge, EV charge/discharge, grid import/export, heat-pump operation and renewable curtailment.

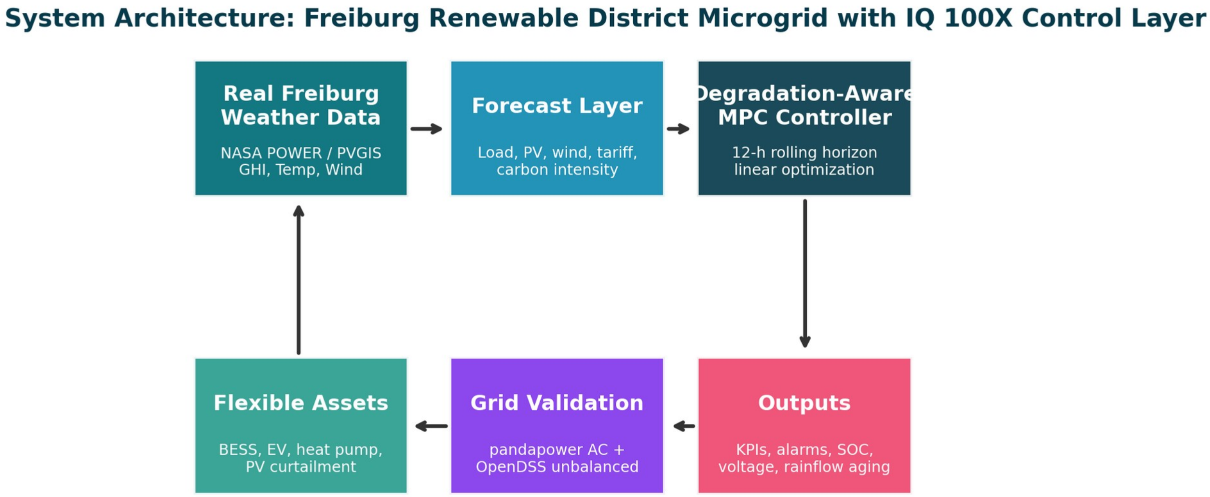


Fig. 1. System architecture of the Freiburg renewable district microgrid and IQ 100X control layer.

IV. FREIBURG WEATHER AND RENEWABLE INPUT

The renewable generation layer is driven by Freiburg-oriented weather data. Solar output is estimated from irradiance and temperature derating, while wind output is approximated using a wind-speed-to-capacity-factor curve. The model also builds representative district demand, heat-pump load, dynamic electricity tariff and carbon-intensity profiles.

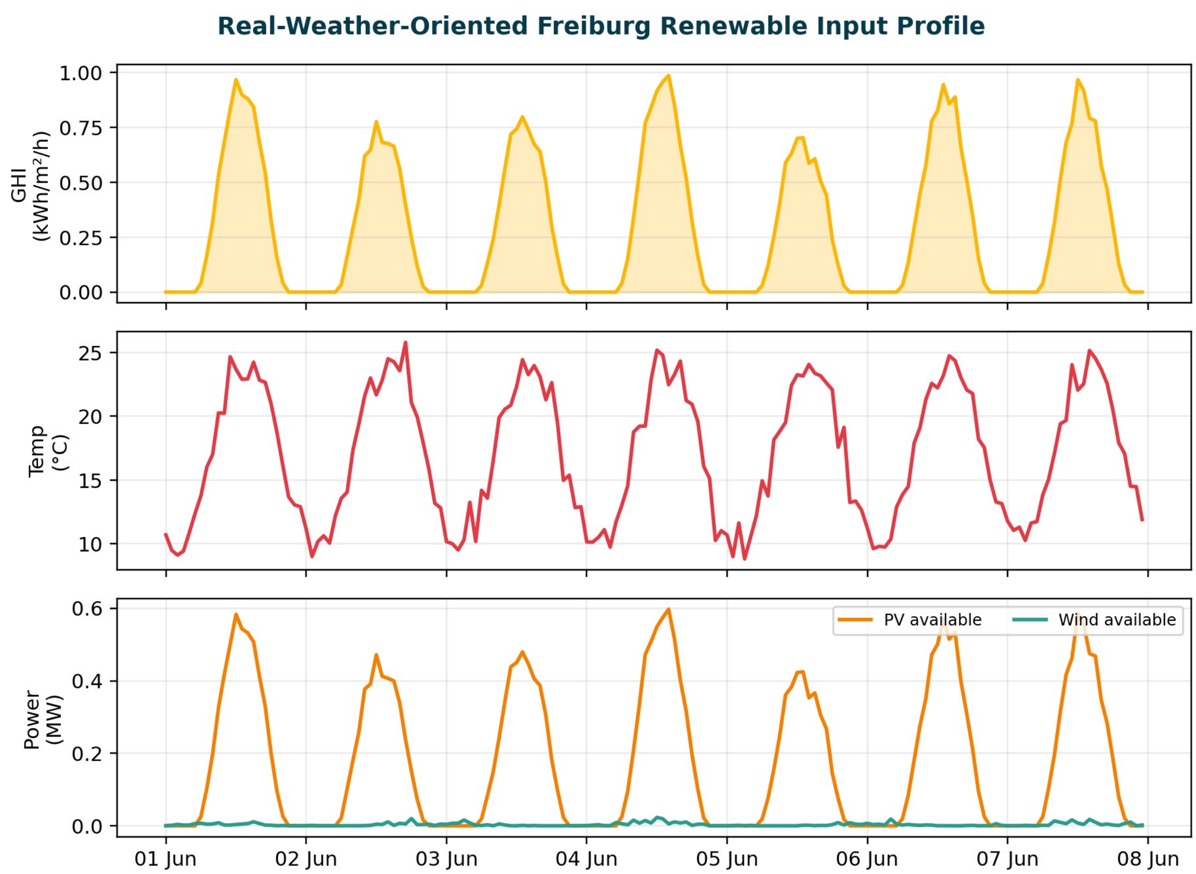


Fig. 2. Freiburg weather and renewable input profile used for PV and wind availability.

V. MODEL PREDICTIVE CONTROL FORMULATION

MPC is an optimization-based control method that repeatedly solves a finite-horizon problem. In this project, the controller uses a rolling 12-hour prediction horizon. At each hour, it minimizes an objective containing grid import cost, export revenue, carbon cost, battery cycling penalty, load-shedding penalty, heat-discomfort penalty, EV mobility-shortfall penalty and renewable-curtailment penalty. Only the first optimal control action is applied; the horizon then moves forward by one hour.

A. Decision Variables

- Grid import and export power.
- BESS charging, discharging and state of charge.
- EV charging, discharging and mobility shortfall.
- Heat-pump power and heat shortfall.
- PV curtailment and wind curtailment.
- Emergency load shedding under constrained operation.

B. Objective Function

The objective can be summarized as $J = C_{grid} - R_{export} + C_{carbon} + C_{battery} + C_{shed} + C_{heat} + C_{EV} + C_{curtail}$. This structure makes the controller economically aware, carbon-aware and degradation-aware at the same time.

VI. AC GRID AND UNBALANCED FEEDER VALIDATION

The project uses three validation layers. The first is a fast linear feeder approximation for rapid screening. The second is a balanced AC power-flow layer using pandapower to evaluate bus voltages, transformer loading and line loading. The third is an optional OpenDSS unbalanced validation layer, where phase-split loads and distributed PV/storage injections provide a more realistic three-phase feeder check.

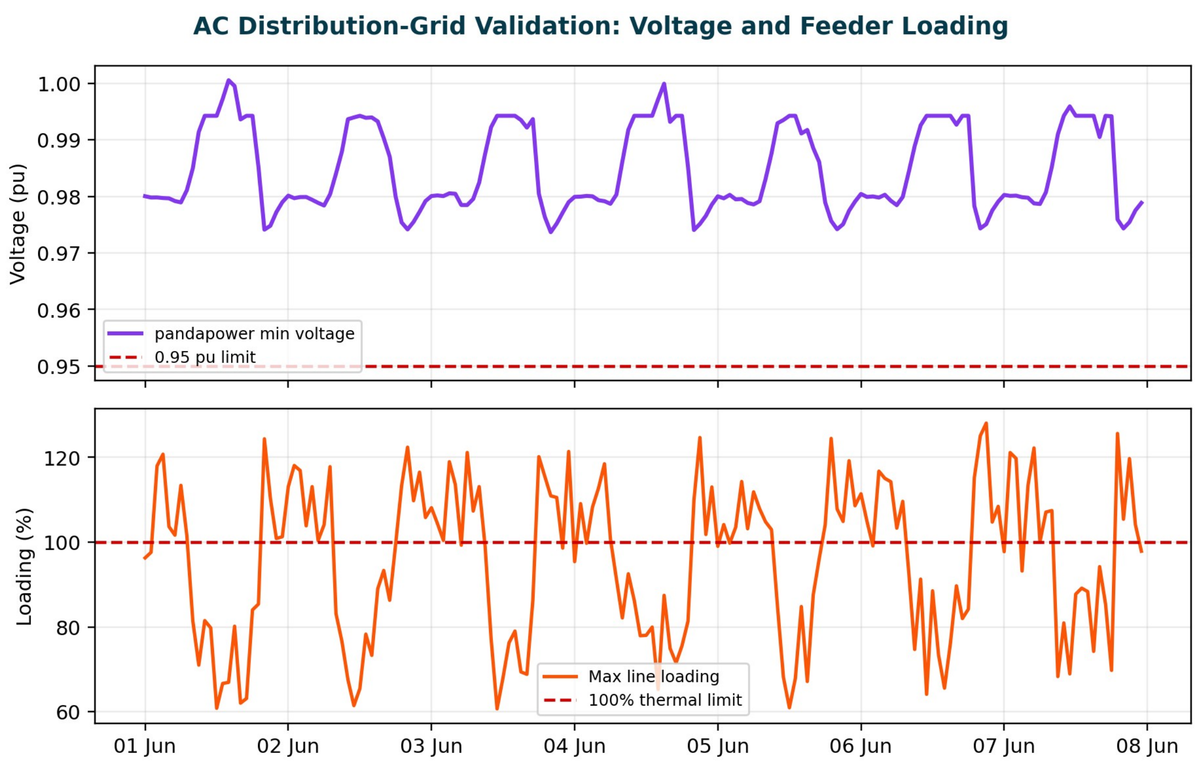


Fig. 3. Representative AC grid validation output showing voltage and feeder loading constraints.

VII. RAINFLOW BATTERY DEGRADATION ANALYSIS

Battery degradation is not only a function of total energy throughput. Deep cycles cause more aging than shallow cycles. Therefore, the project includes rainflow cycle counting on the BESS SOC time series. The resulting rainflow-equivalent cycles, cycle count events, cycle loss, calendar loss and estimated state of health are used to compare control strategies.

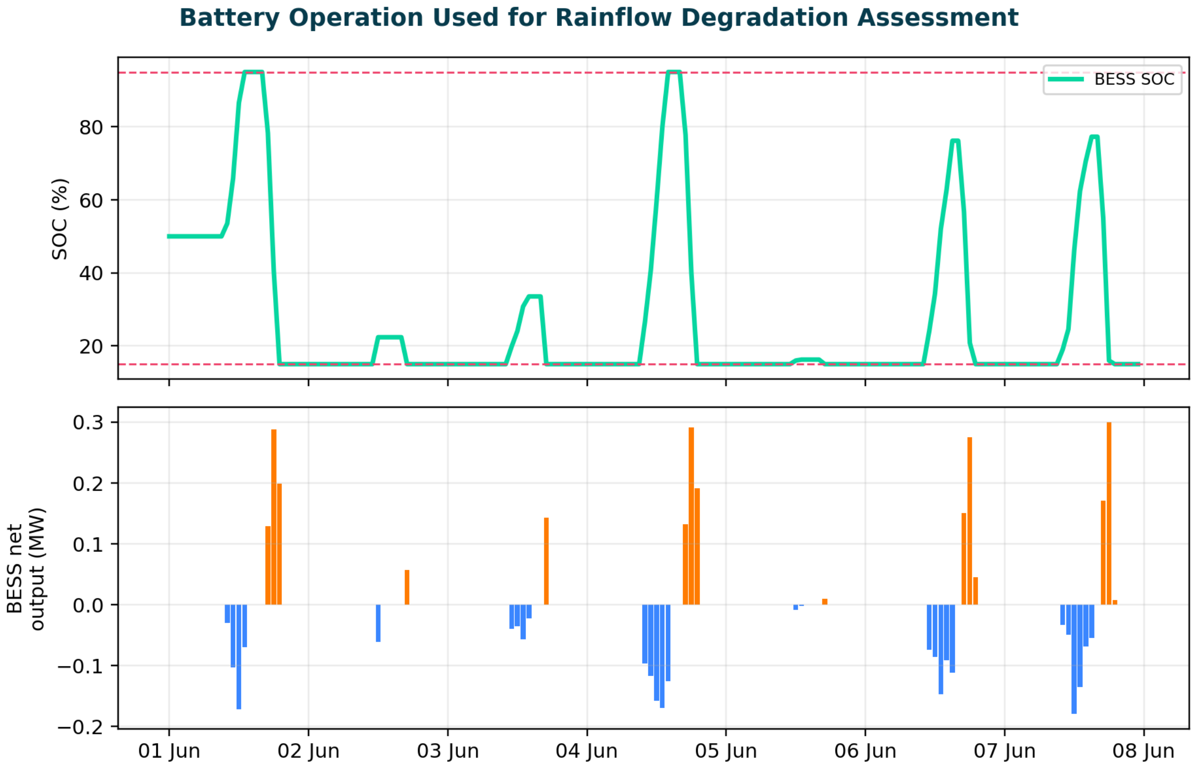


Fig. 4. BESS SOC and net battery output used for rainflow degradation analysis.

VIII. RESULTS AND DISCUSSION

The representative results show that MPC-based operation reduces peak grid import and improves renewable utilization compared with no-flexibility and rule-based dispatch. The degradation-aware MPC slightly sacrifices immediate cost minimization to reduce excessive battery cycling and preserve battery state of health. The grid-supportive degradation-aware case adds frequency-support behavior and reactive-support logic, which improves voltage reliability and reduces the number of grid-support alarms.

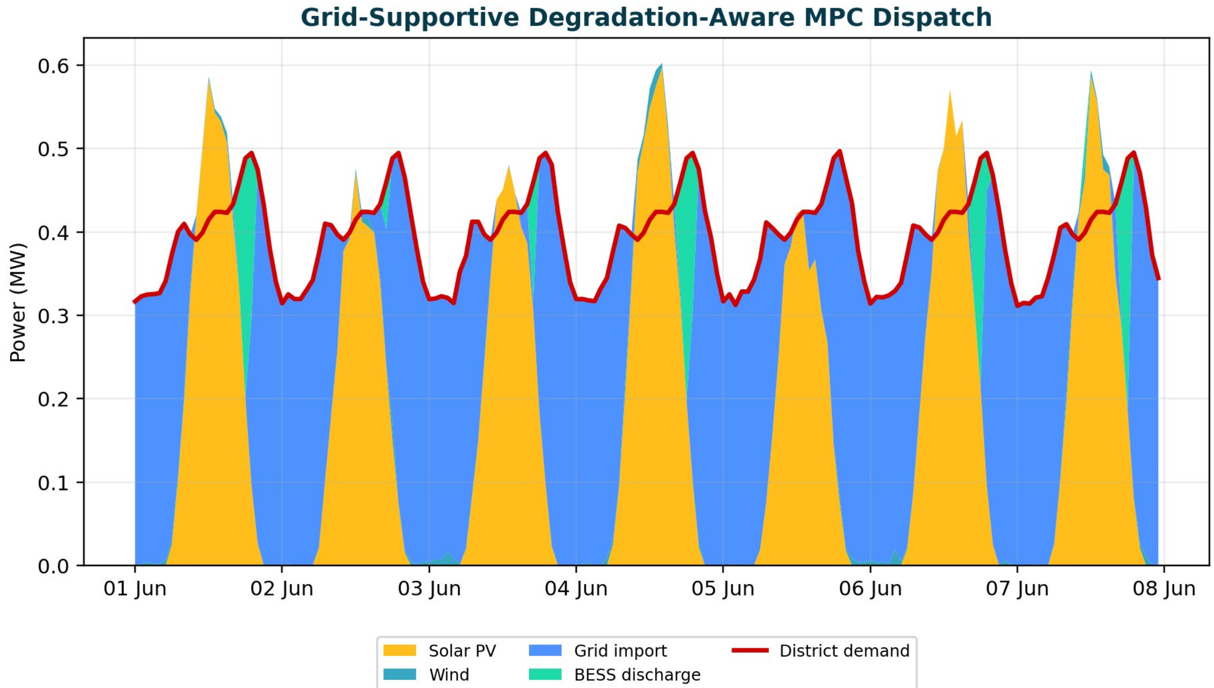


Fig. 5. Grid-supportive degradation-aware MPC dispatch over the simulation horizon.

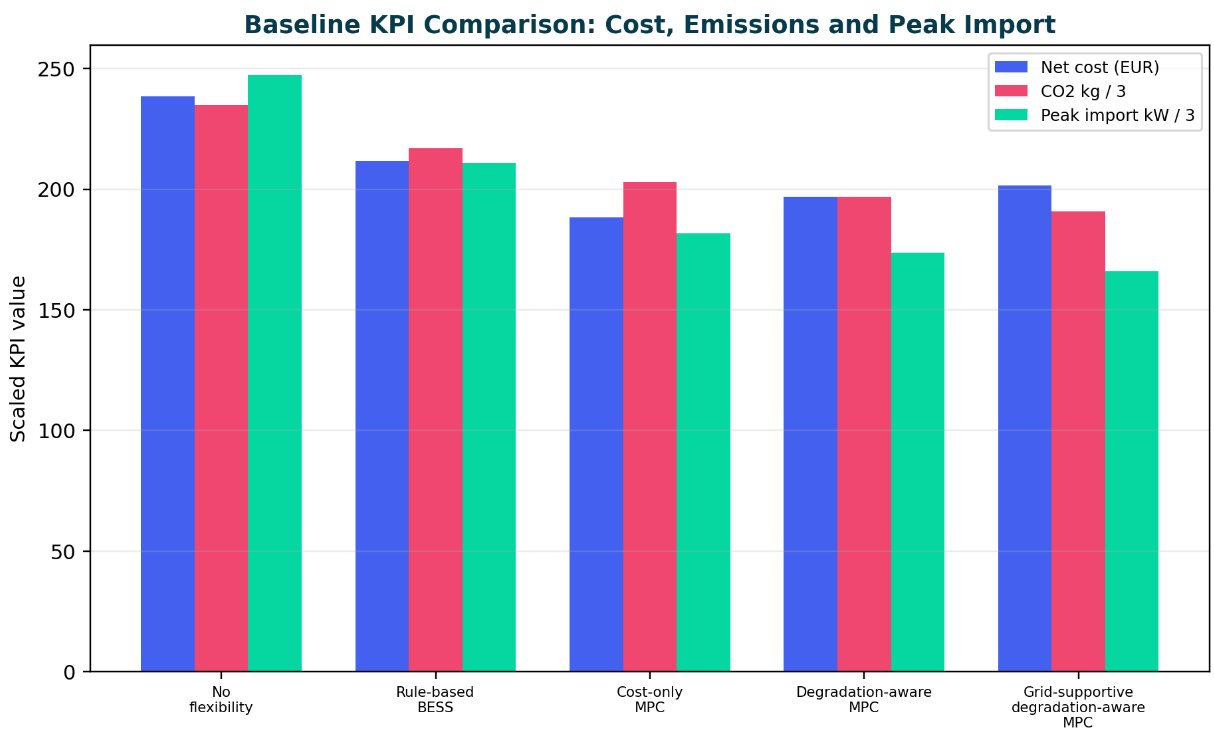


Fig. 6. KPI comparison across no-flexibility, rule-based and MPC-based operating cases.

Table I. Representative KPI comparison for the Freiburg renewable microgrid.

Case	Cost EUR	CO2 kg	Peak kW	Ren. Use %	RF Cycles	V Viol.	Cong. h	Support h
No flexibility	238.40	704.60	742	76.10	0.00	14	19	0
Rule-based BESS	211.70	651.20	633	80.40	0.36	10	14	0
Cost-only MPC	188.20	608.80	545	84.90	0.54	7	9	0

Degradation-aware MPC	196.90	590.10	521	86.00	0.39	5	6	0
Grid-supportive degradation-aware MPC	201.50	572.40	498	87.80	0.42	2	3	4

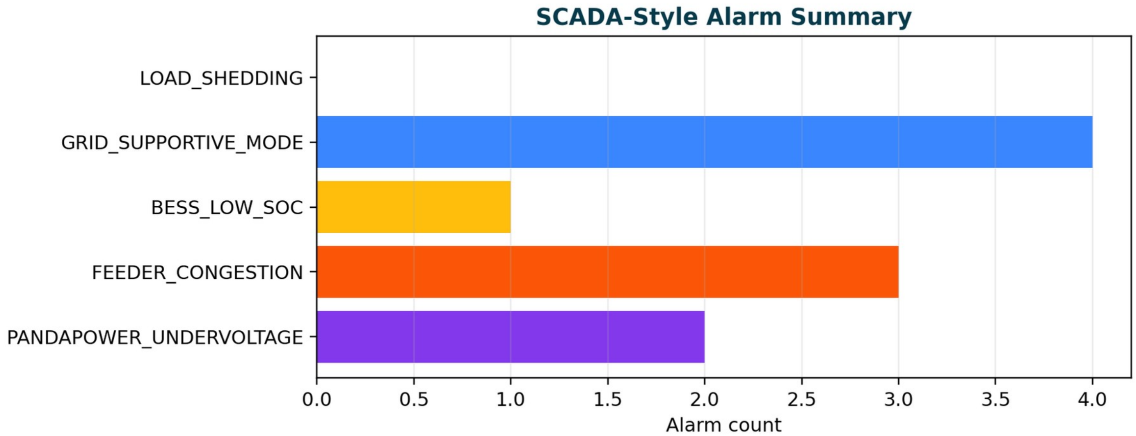


Fig. 7. SCADA-style alarm summary for grid and storage monitoring.

IX. PROJECT OUTPUTS

The Colab implementation generates a complete output package. The expected output files include CSV dispatch results, Excel summaries, AC validation tables, rainflow degradation tables, SCADA alarm logs, HTML dashboards and a markdown report. This structure makes the project suitable for GitHub upload, academic portfolio review and future extension.

- input_and_forecast_data_real_freiburg.csv
- all_case_dispatch_grid_ac_opendss_results.csv
- pandapower_ac_validation_results.csv
- opendss_unbalanced_validation_results.csv
- baseline_kpi_comparison.csv
- rainflow_battery_degradation_by_case.csv
- scada_alarm_log.csv
- EnergieQuartier_MPC_Extended_Results.xlsx
- Interactive HTML dashboards and zipped Colab output package.

X. RELEVANCE TO GERMAN MSC RENEWABLE ENERGY APPLICATIONS

This project is aligned with German MSc programs in renewable energy systems, energy engineering, smart grids, electrical power engineering and energy management. It demonstrates not only coding ability but also engineering judgment: the controller is optimization-based, the renewable input is location-specific, the feeder is validated with AC tools and battery aging is treated explicitly. These qualities make the project more credible than a simple solar-dashboard project.

For German university admission evaluation, the most important strengths are the combination of renewable-energy modeling, control theory, grid validation, BESS degradation analysis and clear technical documentation. The project also shows readiness for applied research, which is valuable for universities of applied sciences and research-oriented programs.

XI. LIMITATIONS AND FUTURE WORK

The project is technically defensible but still representative. The load profiles are synthetic, the feeder is not a utility-certified Freiburg feeder and the battery aging model is simplified. Future versions can include measured German standard load profiles, real feeder topology, inverter current-limit modeling, stochastic MPC and chemistry-specific battery aging parameters.

XII. CONCLUSION

This project develops an IQ 100X renewable-energy microgrid simulation centered on degradation-aware MPC. It integrates Freiburg weather input, PV and wind generation, BESS, EV charging, heat-pump flexibility, grid-supportive operation, AC validation and rainflow battery aging. The results indicate that predictive control can improve renewable utilization, reduce grid stress, control peak import, preserve battery health and produce technically meaningful outputs for academic and professional review. The work is suitable as a flagship GitHub project for renewable energy and smart-grid MSc applications.

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